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RESOLUTION MSC.524(106) (adopted on 10 November 2022)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)

RESOLUTION MSC.524(106)
(adopted on 10 November 2022)

AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

NOTING resolution MSC.391(95), by which it adopted the International Code of Safety for Ships using Gases or other Low -flashpoint Fuels (IGF Code), which has become mandatory under chapters II-1 and II-2 of the International Convention for the Safety of Life at Sea, 1974 ("the Convention"),

NOTING ALSO article VIII(b) and regulation II -1/2.29 of the Convention concerning the procedure for amending the IGF Code,
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NOTING ALSO article VIII(b) and regulation II -1/2.29 of the Convention concerning the procedure for amending the IGF Code,

HAVING CONSIDERED, at its 106th session, amendments to the IGF Code proposed and circulated in accordance with article VIII(b)(i) of the Convention:

1 ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IGF Code, the text of which is set out in the annex to the present resolution;

2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2025, unless, prior to that date, more than one -third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet have notified their objections to the amendments;

3 INVITES Contracting Governments to the Convention to note that, in accordance with
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3 INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall ente r into force on 1 January 2026 upon their acceptance in accordance with paragraph 2 above;

4 REQUESTS the Secretary -General, for the purposes of article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of t he amendments contained in the annex to all Contracting Governments to the Convention;

5 ALSO REQUESTS the Secretary-General to transmit copies of this resolution and its annex to Members of the Organization which are not Contracting Governments to the Convention.

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ANNEX
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
PART A-1

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ANNEX
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
PART A-1
SPECIFIC REQUIREMENTS FOR SHIPS USING NATURAL GAS AS FUEL
7 Material and general pipe design
7.4 Regulations for materials
7.4.1 Metallic materials
Table 7.3 is replaced in its entirety, by the following:
"Table 7.3
PLATES, SECTIONS AND FORGINGS see note 1 FOR FUEL TANKS, SECONDARY
BARRIERS AND PROCESS PRESSURE VESSELS FOR DESIGN TEMPERATURES
BELOW MINUS 55°C AND DOWN TO MINUS 165°C see note 2
Maximum thickness 25 mm see notes 3 and 4
Minimum design
temp. (°C) Chemical composition see note 5 and heat treatment Impact test temp. (°C)
-60 1.5% nickel steel normalized or normalized and tempered or quenched and tempered or TMCP see note 6 -65
-65
2.25% nickel steel normalized or normalized and
tempered or quenched and tempered or TMCP
see notes 6 and 7
-70
-90
3.5% nickel steel normalized or normalized and

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tempered or quenched and tempered or TMCP
see notes 6 and 7
-70
-90
3.5% nickel steel normalized or normalized and
tempered or quenched and tempered or TMCP
see notes 6 and 7
-95
-105 5% nickel steel normalized or normalized and tempered or quenched and tempered see notes 6, 7 and 8 -110
-165 9% nickel steel double nor malized and tempered or quenched and tempered see note 6 -196
-165 Austenitic steels, such as types 304, 304L, 316, 316L, 321 and 347 solution treated see note 9 -196
-165 High manganese austenitic steel hot rolling and controlled cooling see notes 10 and 11 -196
-165 Aluminium alloys, such as type 5083 annealed Not required
-165 Austenitic Fe -Ni alloy (36% nickel). Heat treatment as agreed Not required
TENSILE AND TOUGHNESS (IMPACT) TEST REGULATIONS
Sampling frequency
 Plates Each "piece" to be tested
 Sections and forgings Each "batch" to be tested
Toughness (Charpy V-notch test)

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Sampling frequency
Plates Each "piece" to be tested
Sections and forgings Each "batch" to be tested
Toughness (Charpy V-notch test)
Plates Transverse test pieces. Minimum average energy value (KV) 27J
Sections and forgings Longitudinal test pieces. Minimum average energy (KV) 41J

Notes

1. The impact test required for forgings used in critical applications shall be subject to special consideration by the Administration.

2. The regulations for design temperatures below -165°C shall be specially agreed with the Administration.

3. For materials 1.5% Ni, 2.25% Ni, 3.5% Ni and 5% Ni, with thicknesses greater than 25 mm, the impact tests shall be conducted as follows:

Material thickness (mm) Test temperature (°C)
25 < t 30 10°C below design temperature

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25 mm, the impact tests shall be conducted as follows:

Material thickness (mm)	Test temperature (°C)
25 < t 30	10°C below design temperature
30 < t 35	15°C below design temperature
35 < t 40	20°C below design temperature

The energy value shall be in accordance with the table for the applicable type of test specimen. For material thickness of more than 40 mm, the Charpy V -notch values shall be specially considered.

4. For 9% Ni stee ls, austenitic stainless steels, high manganese austenitic steels and aluminium alloys, thickness greater than 25 mm may be used.

5. The chemical composition limits shall be in accordance with recognized standards.

6. Thermo-mechanical controlled process ing (TMCP) nick el steels will be subject to acceptance by the Administration.

7. A lower minimum design temperature for quenched and tempered steels may be specially agreed with the Administration.

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7. A lower minimum design temperature for quenched and tempered steels may be specially agreed with the Administration.

8. A specially heat -treated 5% nickel steel, for example triple heat -treated 5% nickel steel, may be used down to -165°C, provided that the impact tests are carried out at -196°C.

9. The impact test may be omitted subject to agreement with the Administration.

10. The use of the material shall be subject to th e required cond itions specified by the Administration based on the Guidelines developed by the Organization.

11. The impact test may not be omitted for high manganese austenitic steel."

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